

TUTORIAL – 13

Selenium Environment Setup

Aim:

To create and configure Python, Selenium WebDriver, PyCharm IDE, browser drivers and automation testing libraries for Selenium testing.

Description:

Selenium automation requires a programming environment, Selenium library, browser and WebDriver.

In this practical, students will configure the complete Selenium environment in **PyCharm** and verify the configuration by executing a basic Selenium program.

Problem Statement:

Create a Selenium automation environment using:

- Python
- PyCharm
- Selenium WebDriver
- Google Chrome

Execute a Python script that launches Chrome and opens a web page.

Implementation Details

Parameter	Details
Project Name	Selenium Environment Setup
Application/Software	Selenium WebDriver
Module Name	Environment Configuration
SDLC Model	Agile
Programming Language	Python
IDE/Tool Used	PyCharm
Input/Test Data	Website URL

Procedure – Complete PyCharm Setup

Step 1 – Open PyCharm

Open PyCharm.

Select:

New Project → SoftwareTesting

Step 2 – Select Python Interpreter

Select a Python interpreter available on the computer.

Step 3 – Create Python File

Create:

tutorial13.py

Step 4 – Install Selenium

Open:

PyCharm → Terminal

Execute:

pip install selenium

Step 5 – Verify Selenium

Execute:

pip show selenium

Step 6 – Write Python Program

```
from selenium import webdriver
```

```
driver = webdriver.Chrome()
```

```
driver.maximize_window()
```

```
driver.get("https://www.selenium.dev/")
```

```
print("Browser launched successfully")
```

```
print("Page Title:", driver.title)
```

```
print("Current URL:", driver.current_url)
```

```
driver.quit()
```

Step 7 – Run Program

Right-click:

tutorial13.py

Select:

Run 'tutorial13'

Step 8 – Observe Browser

Google Chrome should open automatically and load the Selenium website.

Step 9 – Observe PyCharm Console

Expected output will show:

Browser launched successfully

Page Title: Selenium

Current URL: <https://www.selenium.dev/>

SDLC–STLC Mapping Table

SDLC Phase	Corresponding STLC Phase	Objective	Key Activities
Requirement Analysis	Requirement Analysis	Identify setup requirements	Identify tools
System Design	Test Planning	Plan environment	Select browser and IDE
Development	Test Environment Setup	Configure tools	Install Python/Selenium
Testing	Test Execution	Verify configuration	Execute Selenium script
Deployment	Test Closure	Confirm environment	Record result
Maintenance	Maintenance Testing	Maintain environment	Update libraries

Source Code

```
from selenium import webdriver
```

```
driver = webdriver.Chrome()

driver.maximize_window()

driver.get("https://www.selenium.dev/")

print("Browser launched successfully")
print("Page Title:", driver.title)
print("Current URL:", driver.current_url)

driver.quit()
```

Test Case Detailed

Field	Details
Test Case ID	TC-13-01
Requirement ID	REQ-13-01
Module Name	Selenium Environment
Feature Under Test	Browser Launch
Test Scenario	Launch Chrome using Selenium
Test Objective	Verify Selenium configuration
Test Type	System
Test Technique	Black Box
Priority	High
Severity	High
Preconditions	Python, PyCharm and Chrome installed
Postconditions	Browser closed successfully
Test Environment	Windows + Python + PyCharm
Browser/OS	Chrome / Windows

Test Data	Selenium website URL
Expected Result	Chrome launches and website opens
Actual Result	Chrome launched and website opened
Status	Pass
Executed By	_____
Execution Date	_____

Test Case Execution

Step No.	Test Steps	Input	Expected Result	Actual Result	Status
1	Open PyCharm	PyCharm	IDE opens	Opened	Pass
2	Check Python	Python Interpreter	Python available	Available	Pass
3	Install Selenium	pip install selenium	Selenium installed	Installed	Pass
4	Run script	tutorial13.py	Chrome launches	Launched	Pass
5	Validate website	Selenium URL	Website opens	Opened	Pass

Testing Activities Performed

Sr. No.	SDLC Phase	STLC Phase	Testing Activity	Deliverable/Artifact
1	Requirement Analysis	Requirement Analysis	Identified setup requirements	Requirement List
2	System Design	Test Planning	Planned test environment	Environment Plan
3	Development	Environment Setup	Installed/configured tools	Configured Environment
4	Testing	Test Execution	Executed Selenium program	Test Result

5	Deployment	Test Closure	Verified setup	Test Summary
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Project Deliverables Table

SDLC Phase	Deliverable / Artifact	Prepared By
Requirement Analysis	SRS	Business Analyst
System Design	Test Design	System Analyst
Development	Source Code	Developer
Testing	Test Cases/Test Report	Tester
Deployment	Test Summary Report	QA
Maintenance	Regression Report	Tester

Roles & Responsibilities

Role	Responsibility	SDLC/STLC Phase
Project Manager	Manage project	All
Business Analyst	Define requirements	Requirement
Software Developer	Develop software	Development
Software Tester	Execute tests	Testing
Client / End User	Validate application	UAT

Project Documents

Sr. No.	Document / Artifact	Purpose	Status
1	Software Requirement Specification	Defines software requirements	✓
2	Test Plan	Planning testing activities	✓
3	Test Cases	Defines test scenarios	✓
4	Test Data	Input used for testing	✓
5	Bug Report	Records identified defects	✓

6	Test Summary Report	Summarizes testing results	✓
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Observation

Sr. No.	Observation	Remarks
1	Python environment configured	Successful
2	Selenium installed	Successful
3	PyCharm configured	Successful
4	Chrome launched using Selenium	Successful
5	Website opened successfully	Successful

Output

Students should capture screenshots of:

1. PyCharm project
2. Python interpreter
3. Selenium installation
4. Source code
5. Chrome browser
6. PyCharm console output

Questionnaire

Q1. Which library is used for browser automation in Python?

Answer: C) Selenium

Q2. What is the purpose of ChromeDriver?

Answer: B) Browser Communication with Selenium

Q3. Which command is used to install Selenium?

Answer: C) pip install selenium

Q4. Which IDE is used for Selenium script development in this practical?

Answer: B) PyCharm

Q5. What is the purpose of WebDriver?

Answer: Browser control and communication between Selenium and the browser.